

AMENDMENT NO. 1 FEBRUARY 1987
TO
IS : 3203-1982 METHODS FOR TESTING LOCAL
THICKNESS OF ELECTROPLATED COATINGS

(First Revision)

(Page 7, clause 5.1) — Add the following new clause after 5.1:

'5.1.1 Accuracy — Under optimum conditions, the errors involved in this method are normally within ± 15 percent.'

(Page 8, clause 5.3.1.1, line 2) — Substitute the following for the existing matter:

'Anhydrous ferric chloride (FeCl_3) 180 g'

(Page 9, clause 5.3.2.1) — Add the following new clause after 5.3.2.1 :

'5.3.2.2 To distinguish between bright and dull electroplated nickel coatings, a simple method of testing as given in Appendix B may be adopted.'

(Page 10, clause 5.3.6) — Substitute '260 g' for '250 g' against Potassium iodide.

(Page 10, clause 5.3.7, line 1) — Substitute '100 g' for ' 120 g'.

(Page 18, clause 6.3) — Add the following Note under 6.3:

'NOTE — The test specimen and test solution should normally be kept between 16 and 25°C. The temperature up to 30°C is permissible since the measurement becomes less reliable at the higher temperature.'

(Page 18, clause 6.5) — Substitute the following for the existing clause:

'6.5 Record the time to the nearest half second with a stop-watch from the appearance of black film over the surface under test to the cessation of gassing. Note down room temperature near the test area to accuracy of 0.2°C.'

(Page 18, clause 6.6) — Add the following new clause after 6.6:

'6.7 Accuracy -- The thickness determination by this method has an uncertainty about ± 20 percent, and is, therefore, much less accurate than coulometric method specified in the product specifications.'

(Page 19, clauses A-3 and A-3.1, line 1) — Substitute ' coulometric ' for ' colometric '.

(Page 19, clause A-3.2) — Add the following Appendix B after this clause:

‘APPENDIX B

(Clause 5.3.2.2)

METHOD OF TESTING TO DISTINGUISH BETWEEN BRIGHT AND DULL ELECTROPLATED NICKEL COATINGS

B-1. GENERAL

B-1.1 In testing the thickness of nickel (single) coatings the usual end points have been given. But the coating sometimes shows a darkening of the test-spot by a non-adherent blackish matter which may need to be wiped off at intervals with a stub of paper. While the dull and cobalt-nickel bright-nickel coatings give a normal rate of attack and no correction is needed, the rate of penetration of 'organic bright-nickel' coatings may be different and a correction factor may be required. These organic bright-nickel coatings may be detected as given below.

B-2. PREPARATION OF SURFACE

B-2.1 Roughen the surface slightly by rubbing with fine pumice powder (no other abrasive power should be used). Dry the surface with a clean cloth.

B-3. PREPARATION OF THE TEST SOLUTION

B-3.1 The test solution shall be prepared by dissolving 20 g of chromic acid and 10 ml of concentrated sulphuric acid in 100 ml of distilled water.

B-4. PROCEDURE

B-4.1 Apply a drop of cold test solution on the clean surface. If attack occurs rendering the drop brown within 5 minutes, the deposit is organic bright-nickel.

NOTE — It is reported that organic bright-nickel deposits from solutions containing sulphonic acids and other organic additives dissolve more rapidly than dull-nickel and that the true thickness may be obtained by multiplying the jet-test thickness by a factor of 1.5. However, this factor should be obtained from the manufacturer of the proprietary brighteners, as this factor may vary from one manufacturer to another.

B-4.2 Correction Factor — Correction curves based to proprietary nickel brighteners and an average curve is given in Fig. 9. The true thickness of bright nickel depends on the nickel thickness as calculated from Fig. 2 of this standard. In case of doubt it is advisable to compare jet-test results on 3 or 4 samples of different nickel thicknesses with thickness measurement by some other standard methods to see if the curve is needed or a simple correction will suffice.'

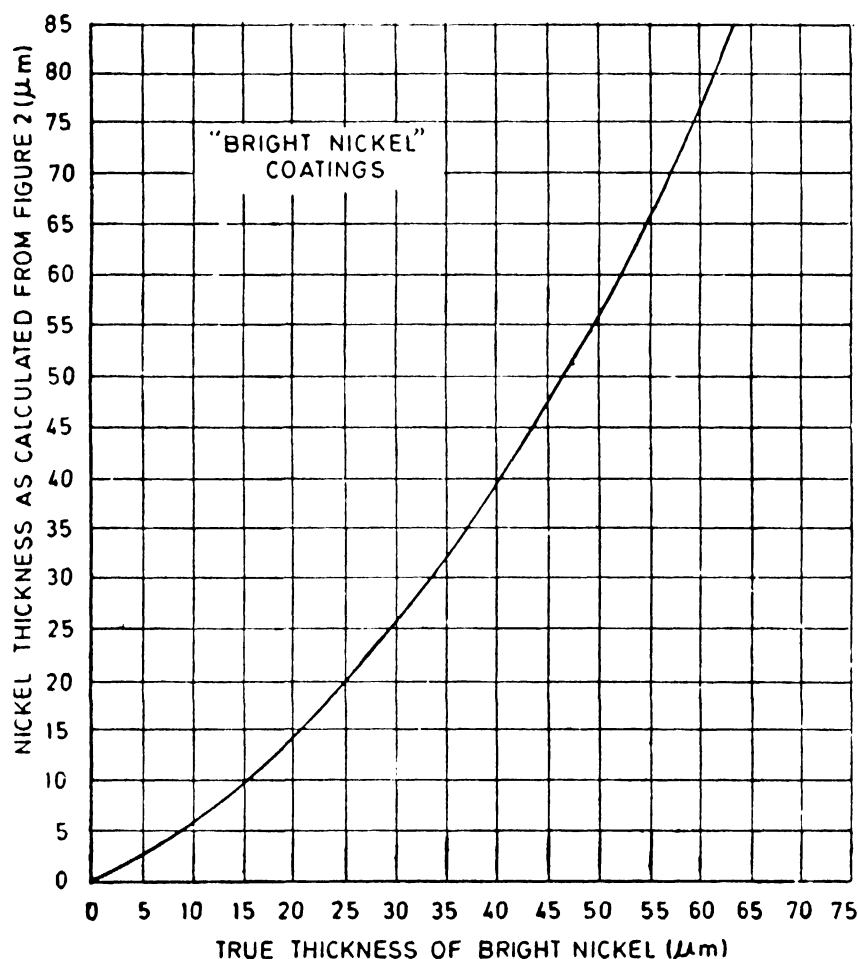


FIG. 9 CORRECTION CURVES BASED TO PROPRIETARY NICKEL BRIGHTENERS AND AN AVERAGE CURVE

(SMDC 23)